

Initialize Parameters for sim

```
FFT_LENGTH = 512;  
specAvg = 27;
```

Filter valid data

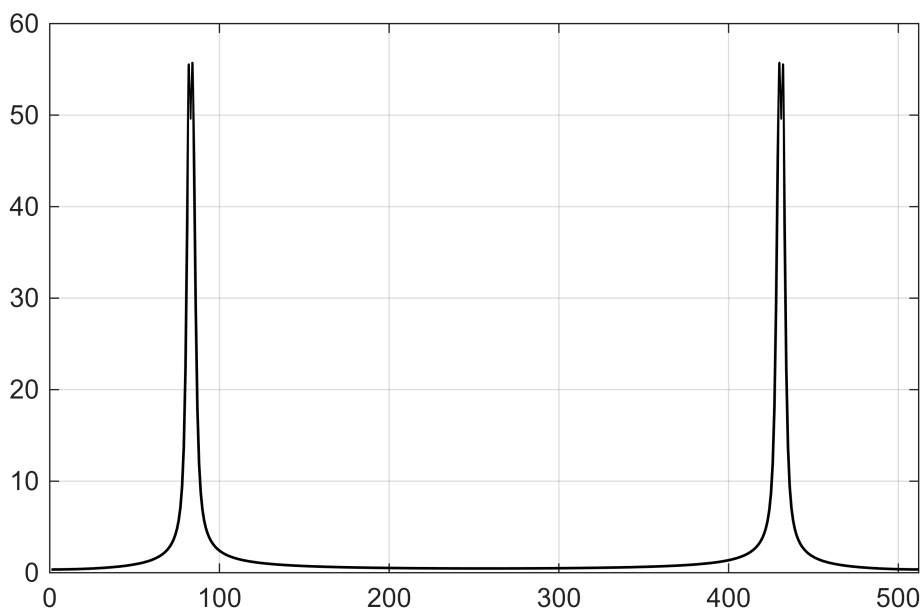
```
% k = find(out.SciInfo_Valid.Data, 1);  
% fftData = out.SciInfo.Data(k:end);  
k = find(out.mag_valid, 1)
```

```
k =  
1118
```

```
fftData = out.mag_out(k:end);  
fftData = reshape(fftData, [], 1); % Ensure fftData is 2D for plotting later
```

Arrange data in columns with 512 (FFT_LENGTH) Rows

```
a = 1:FFT_LENGTH;  
a = a';  
% subplot(2, 1, 1)  
figure;  
plot(a, fftData(1:FFT_LENGTH), 'k', 'LineWidth', 1)  
xlim([0 FFT_LENGTH])  
grid on;
```



Find number of spectral averages

```
% specAvg = ceil(length(fftData)/FFT_LENGTH/2)
```

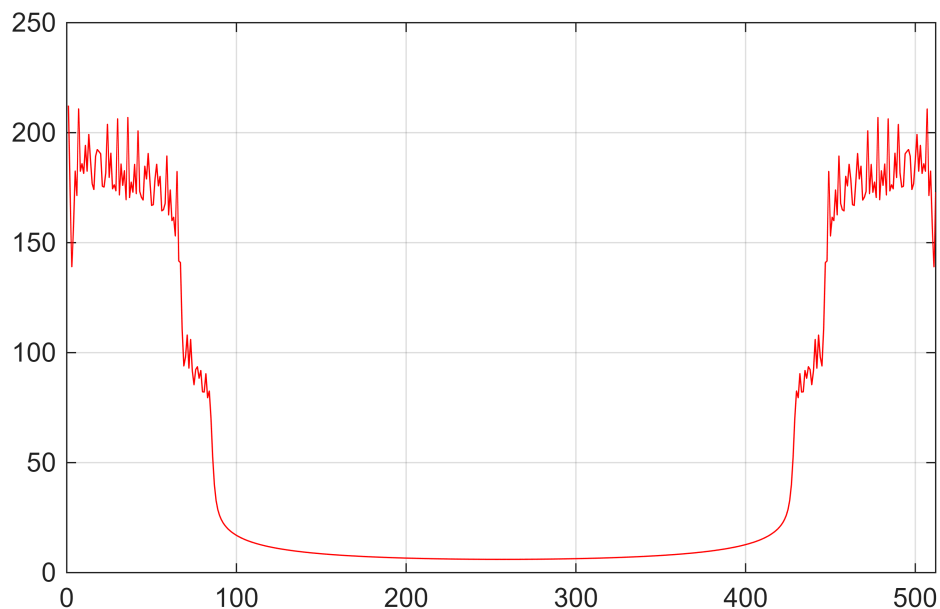
```
specAvg = floor(length(fftData)/FFT_LENGTH) - 1
```

```
specAvg =  
26
```

```
% specAvg = 30  
fftData_snr1 = fftData(1:specAvg*FFT_LENGTH);  
fftData_snr2 = reshape(fftData_snr1, FFT_LENGTH, specAvg);
```

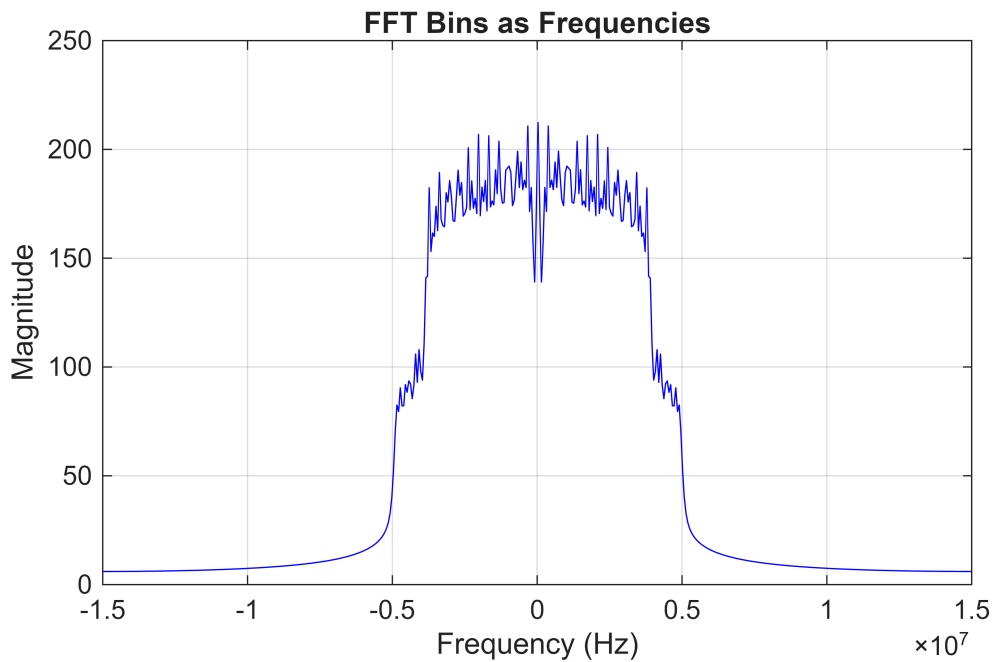
Sum along columns and plot

```
fftData_snr3 = sum(fftData_snr2, 2);  
  
% subplot(2, 1, 2)  
figure;  
plot(a, fftData_snr3, 'r')  
xlim([0 FFT_LENGTH])  
grid on;
```



Plot data over frequency

```
frequencies = linspace(-Fs/4, Fs/4, FFT_LENGTH); % /4 because this is downsampled  
figure;  
plot(frequencies, fftshift(fftData_snr3), 'b')  
xlabel('Frequency (Hz)')  
ylabel('Magnitude')  
title('FFT Bins as Frequencies')  
xlim([-Fs/4 Fs/4])  
grid on;
```



Plot Simulink Results

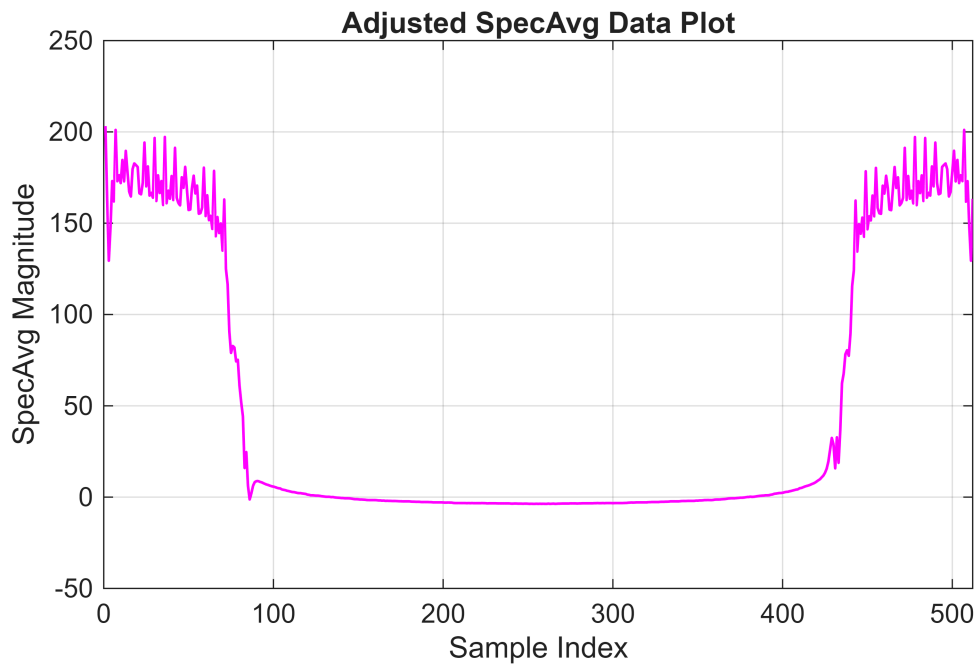
```
% % Plot the Simulink results from out.SpecAvg and out.SpecAvg_Valid
validIndices = find(out.avg_valid, 1)
```

```
validIndices =
14430
```

```
specAvgData = out.avg_out(validIndices:end);
specAvgData = reshape(specAvgData, [], 1); % Ensure specAvgData is 2D for plotting
later
```

```
% Ensure specAvgData is of appropriate size for plotting
% Allows plotting partial FFTs if specAvg is too large
if length(specAvgData) < FFT_LENGTH
    specAvgData = [specAvgData; zeros(FFT_LENGTH - length(specAvgData), 1)]; % Pad
with zeros if necessary
else
    specAvgData = specAvgData(1:FFT_LENGTH); % Truncate if too long
end
```

```
figure;
plot(a, specAvgData, 'm', 'LineWidth', 1)
xlim([0 FFT_LENGTH])
grid on;
xlabel('Sample Index')
ylabel('SpecAvg Magnitude')
title('Adjusted SpecAvg Data Plot')
```



```
% Plot the Simulink results over the frequency axis
figure;
plot(frequencies, fftshift(specAvgData), 'r')
xlabel('Frequency (Hz)')
ylabel('SpecAvg Magnitude')
title('SpecAvg Data over Frequency')
xlim([-Fs/4 Fs/4])
grid on;
```

